# DSP in VLSI
# Homework 4 - CORDIC

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1. (Step 1) Please draw the figure showing $S(N)$ versus $N$ for $N = 1 \ldots 30$. (5%)

   Figure 1 shows the CORDIC scaling factor $S(N) = \prod_{i=0}^{N-1} \frac{1}{\sqrt{1+2^{-2i}}}$ versus the number of micro-rotations $N$. The factor decreases rapidly and converges to approximately 0.6073 by $N \approx 5$, remaining constant thereafter.

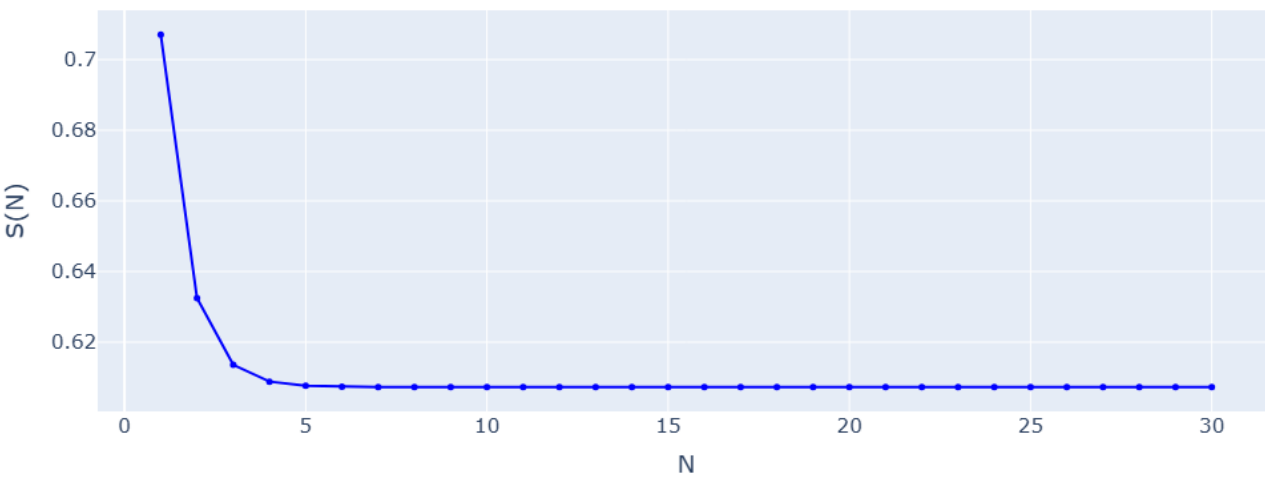


Figure 1: CORDIC scaling factor $S(N)$ versus $N$

2. (Step 2) Draw the figure of average absolute error versus fractional wordlength (10%) to show how you determine the setting of word-length of the fractional part. Write down the integer word-length of $X(i)$ and $Y(i)$ that you use all the stages. Please explain it. (5%)

   Figure 2 plots the average absolute phase error against the fractional wordlength $w$ with $N = 30$ micro-rotations. The error decreases monotonically and falls below the $2^{-9}$ criterion at $w = 9$, after which further increase yields negligible improvement. Therefore, the fractional wordlength of the $X/Y$ datapath is set to $w = 9$ bits.

   Each CORDIC micro-rotation is a pseudo-rotation that does not preserve the vector magnitude; the amplitude grows by a factor of $1/S(N)$. For unit-circle inputs ($\sqrt{X_0^2 + Y_0^2} = 1$) the maximum value reached by either register throughout all $N$ stages is bounded by

$$\max_i \sqrt{X(i)^2 + Y(i)^2} \;=\; \frac{1}{S(N)} \;\approx\; 1.647 \;<\; 2^1.$$

   Since the pseudo-magnitude never reaches 2, one integer bit suffices for all stages. Combined with the sign bit, the complete word format is (1 sign + 1 integer + 9 fractional) = 11 bits.

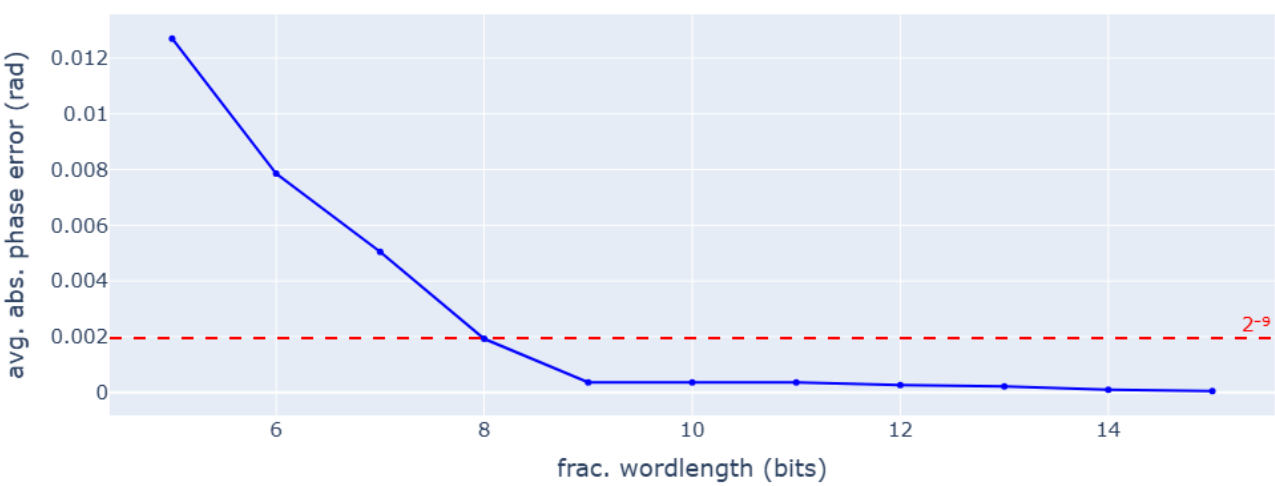


Figure 2: Average absolute phase error versus fractional wordlength $w$ of $X(i)/Y(i)$ ($N = 30$, floating-point $\theta$)

3. (Step 3) Please draw a figure to denote the average phase errors of 10 quantized input pairs $(X, Y)$ versus different numbers of micro-rotations $S$ (10%) and draw a figure to show the resulted phase errors of 10 quantized input pairs versus the word-length of quantized elementary angles (10%). Explain your selection to satisfy the averaged error criterion of $2^{-9}$ radian. Also list a table of the elementary angles (both in floating-point representation and binary fixed-point representation). (5%)

   Figure 3 shows the average absolute phase error versus the number of micro-rotations $S$, with the $X/Y$ fractional wordlength fixed at 9 bits. The error decreases as $S$ increases and first falls below the $2^{-9}$ criterion at $S = 10$. Therefore, the number of micro-rotations is set to $S = 10$.

   Figure 4 shows the average absolute phase error versus the fractional wordlength of the quantized elementary angles, with $S = 10$ micro-rotations and $X/Y$ fractional wordlength fixed at 9 bits. The error decreases with increasing wordlength, meets the $2^{-9}$ criterion at 10 bits, and saturates beyond that point because the dominant error source shifts to the $X/Y$ quantization. Therefore, the fractional wordlength of the elementary angle LUT is set to 10 bits.

   The 10 elementary angles used in the CORDIC micro-rotations are listed in Table 1.

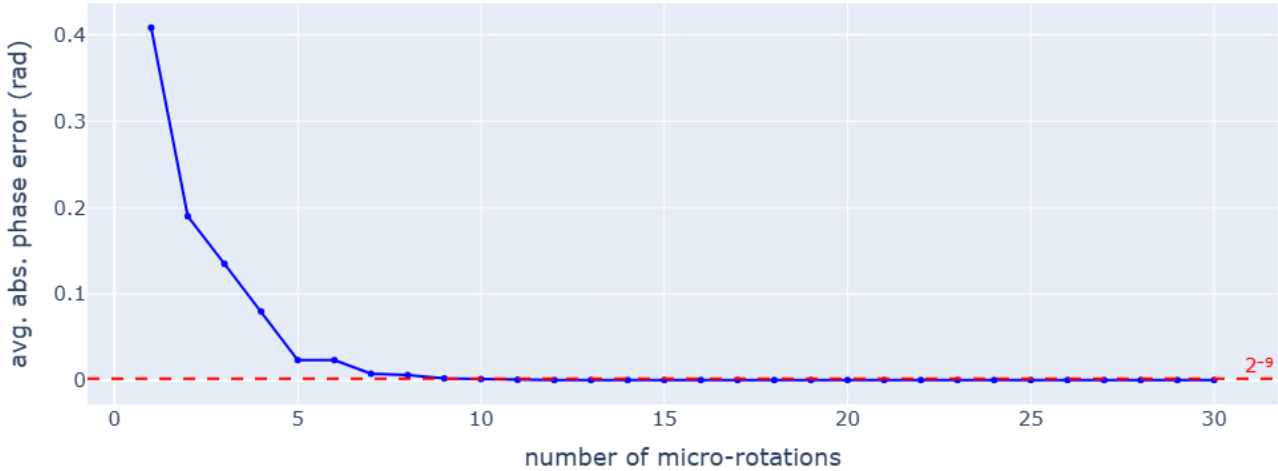


Figure 3: Average absolute phase error versus number of micro-rotations $S$ (fractional wordlength of $X/Y = 9$ bits, floating-point elementary angles)

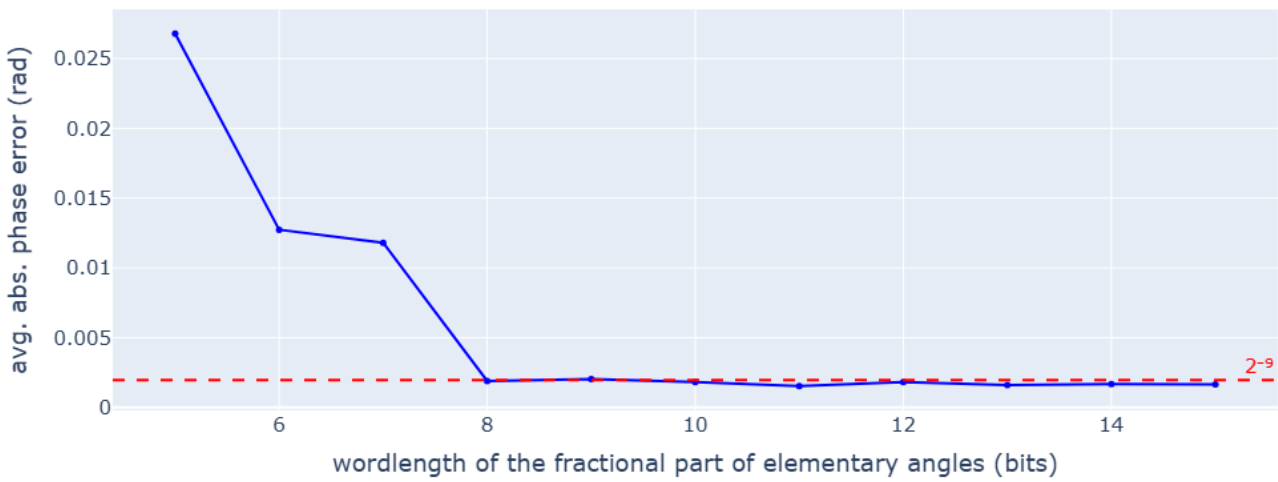


Figure 4: Average absolute phase error versus fractional wordlength of quantized elementary angles ($S = 10$, fractional wordlength of $X/Y = 9$ bits)

| $i$ | $\theta_e(i) = \arctan(2^{-i})$ (rad) | Binary fixed-point (10 frac. bits) |
|---|---|---|
| 0 | 0.7853981634 | `0.1100100100` |
| 1 | 0.4636476090 | `0.0111011011` |
| 2 | 0.2449786631 | `0.0011111011` |
| 3 | 0.1243549945 | `0.0001111111` |
| 4 | 0.0624188100 | `0.0001000000` |
| 5 | 0.0312398334 | `0.0000100000` |
| 6 | 0.0156237286 | `0.0000010000` |
| 7 | 0.0078123411 | `0.0000001000` |
| 8 | 0.0039062301 | `0.0000000100` |
| 9 | 0.0019531225 | `0.0000000010` |

Table 1: Elementary angles $\theta_e(i)$ in floating-point and 10-bit fixed-point binary representation

4. (Step 4) Please show how you decide the number of micro-rotations for the magnitude function with error tolerance of 0.1%. (10%)

Figure 5: Magnitude function error versus number of micro-rotations $N$

5. (Step 5) Write a program to show the setting of fractional wordlength of CSD versus error. Draw the figure. (10%) Depict your design for the shift-and-add block according to your CSD representation. How many adders do you use? (10%)

Figure 6: CSD scaling factor error versus fractional wordlength

Figure 7: Shift-and-add block diagram for the CSD scaling factor

6. (Step 6) Depict your design of the initial stage. Mark the word-length in the block diagram. (10%)

Figure 8: Block diagram of the initial stage with word-lengths marked

7. (Step 6) Implement your design with only DFFs inserted at the inputs and outputs. Show the timing diagram of behavior simulation for four inputs. (15%) Compare the results with the arctangent function and draw the error versus index $m$ to show that your implementation meets the precision requirements of error less than $2^{-9}$. (10%)

Figure 9: Timing diagram of behavior simulation for four inputs

Figure 10: Phase error versus index $m$ for iterative architecture (behavior simulation)

8. (Step 7) Draw the block diagram of your $\frac{S}{2}$-unfolding architecture. (10%) Show the timing diagram of behavior and post-synthesis simulation with correct setting of clock period and 10 inputs. (30%) Verified the error between the post-synthesis results and the floating-point arctangent results. Draw the error versus index $m$. (10%)

Figure 11: Block diagram of the $\frac{S}{2}$-unfolding architecture

Figure 12: Timing diagram of behavior simulation with 10 inputs

Figure 13: Timing diagram of post-synthesis simulation with 10 inputs

Figure 14: Phase error versus index $m$ for $\frac{S}{2}$-unfolding architecture (post-synthesis)

9. (Step 8) Report and compare the area before and after unfolding. (10%)

10. (Step 9) Draw the shift-and-add module for the scaling factor (5%). Connect the shift-and-add module to your micro-rotation module, and calculate the magnitude function. Show the timing diagram of behavior simulation and post-synthesis simulation with proper setting of clock period. (15%) Verified the error between the post-synthesis results and the floating-point arctangent results. Draw the error versus index $m$. (10%)

Figure 15: Complete magnitude function architecture with shift-and-add scaling

Figure 16: Timing diagram of behavior simulation for magnitude function

Figure 17: Timing diagram of post-synthesis simulation for magnitude function

Figure 18: Magnitude error versus index $m$ (post-synthesis)